QRCODE: Quasi-Residual Convex Deep Network for Fusing Misaligned Hyperspectral and Multispectral Images

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***Abstract*— Considering that hyperspectral image (HSI) is often of lower spatial resolution when compared to multispectral image (MSI), an economical approach for obtaining a high-spatial-resolution (HSR) HSI is to fuse the acquired HSI and MSI, thereby greatly facilitating the subsequent material identification and classification in satellite remote sensing. As satellite-acquired HSI and MSI are often misaligned, the proposed deep neural network does not require the input HSI/MSI to be spatially co-registered, making the challenging fusion network design even more difficult. In this study, we propose a streamlined and efficient convex model integrated into the subnetwork, which obviates the need for complex network structures in learning spatial–spectral relationships, effectively guiding the quasi-residual learning task in our alignment-free fusion network. The convex subnetwork is a low-rank model that leverages the convex geometric structure implicitly embedded in the hyperspectral signature space. To address the misalignment between HSI and MSI effectively, we introduce a novel shifted window attention module (SWAM) that exploits the neighboring correlation in the feature domain, significantly enhancing the performance and stability of the fusion task. Capitalizing on the redundancy among spectrums, we employ grouped convolution to decrease the computational complexity without causing additional performance degradation. The proposed quasi-residual convex deep (QRCODE) network demonstrates the state-of-the-art performance in alignment-free HSI/MSI fusion tasks.**

***Index Terms*— Convex optimization (CO), group convolution, hyperspectral image (HSI), image fusion, multispectral image (MSI).**

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I. Introduction

IN THIS era of space exploration, high-spatial-resolution (HSR) hyperspectral image (HSI) is critically important in remote sensing, where the HSR pixels contribute to accurate material classification, while the densely populated spectral bands in the HSI contribute to highly effective material identification [1]. Typically, the spatial resolution of HSI is more limited than that of multispectral image (MSI) [2], [3]. The reason is that hyperspectral imager splits the incident light into more frequency bands, so the spatial region of a single pixel is purposely enlarged for collecting sufficient photons to have a satisfactory signal-to-noise ratio. Due to such a hardware limitation, direct acquisition of HSR HSI is infeasible and less practical. Therefore, a more practical and economical alternative is to separately acquire the HSR MSI and the low-spatial-resolution (LSR) HSI, over the same spatial region under similar atmospheric and illumination conditions; note that acquiring the complementary HSI/MSI becomes increasingly feasible due to the ever-improving revisiting cycles of HSR optical satellites [4]. The acquired HSI and its counterpart HSR MSI are then fused to obtain the desired HSR HSI [5], [6].

With the above research backgrounds, we are ready to recall some existing works (related techniques and current research status) in Section I-A, followed by describing our research content in Section I-B.

A. Existing Works

One category of the fusion methods considers the MSI and HSI as spectrally downsampled and spatially blurred versions of the target image (i.e., the HSR HSI), respectively, and considers the HSI/MSI fusion as an inverse problem for recovering the spatial/spectral details in the target image (denoted by $\boldsymbol{Z}$) from the observable HSI/MSI. The method in [7] models the target image $\boldsymbol{Z}$ using the Gaussian prior, and estimates the associated hyperparameters directly from the data rather than fixing them from the beginning; then, the Gibbs sampler and Hamilton Monte Carlo technique are used to handle the induced complicated joint posterior distribution [8], [9]. Multiresolution analysis can also be adapted for addressing the HSI/MSI fusion problem by linearly combining the MSI bands to synthesize each HSR bands in the target image $\boldsymbol{Z}$ [10], [11], and some related methods are rooted

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in the pansharpening technique [5], [12], [13]. On the other hand, component substitution (CS) is another super-resolution technique that can be used for the HSI/MSI fusion [4], [14].

Another representative fusion approach is based on the hyperspectral unmixing (HU) technique [15], for which the target HSI $\boldsymbol{Z}$ is modeled as $\boldsymbol{Z} = \boldsymbol{AS}$, where $\boldsymbol{A}$ is the materials' hyperspectral signature matrix and $\boldsymbol{S}$ describes the abundances of the materials presented in the scene of interest. In general terms, methods in this category learn the hyperspectral signatures from the observable LSR HSI for recovering $\boldsymbol{A}$, as well as learn the material abundances from the observable HSR MSI for recovering $\boldsymbol{S}$, followed by fusing these information to obtain the desired HSR HSI $\boldsymbol{Z}$ via the low-rank modeling $\boldsymbol{Z} = \boldsymbol{AS}$ [15]. This elegant HU-based idea can be implemented using the so-called coupled non-negative matrix factorization (CNMF) [16], or using its regularized version for better addressing the ill-posed HSI/MSI fusion problem [3]. Nevertheless, all the above-mentioned fusion methods ignore the fact that the acquired counterpart HSI/MSI are often misaligned, making the fusion performance unreliable in practice [3]. As far as we have investigated, the Bayesian inference method (BIM) [17] is more robust against such a misaligned issue [3, Section 4.5]. Such robustness comes from the adoption of overlapping patches (in a roughly estimated target image $\mathbf{Z}$) that are used to learn the overcomplete dictionary, as it has been reported that the patch-overlapping strategy not only reduces block artifacts but improves the robustness against mis-registration [17], [18].

However, Bayesian fusion algorithms are often of high-computational complexity as the involved likelihood or posterior amount to complicated nonconvex objective functions [7]. Therefore, convex analysis-based fusion methods are also quite appealing. As a representative method under the convex analysis framework, hyperspectral super-resolution (HySure) [19] formulates the HSI/MSI fusion as a convex optimization (CO) problem by assuming the availability of $\boldsymbol{A}$ with the unknown $\boldsymbol{S}$ being the optimization variable, regularizes the problem using a convex function known as vector total variation (vecTV) [20], and implements the regularized learning criterion using the CO algorithm called split-augmented Lagrangian shrinkage algorithm (SALSA) [21]. To go a step further, the CO-based CNMF (CO-CNMF) algorithm [3] considers both $\mathbf{A}$ and $\mathbf{S}$ as unknowns, accordingly formulates the fusion problem as a biconvex problem, solves the problem by combining the block coordinate descent method [22] and alternating direction method of multipliers (ADMMs) [23], and guarantees the fusion algorithm to converge to a stationary point. CO-CNMF regularizes the abundances $\boldsymbol{S}$ using the sparsity-promoting ℓ_1-norm, and regularizes the signature matrix $\boldsymbol{A}$ using the so-called volume-demoting (VD) regularizer. VD, rooted in the minimum simplex HU criterion [24], is a convex surrogate of the volume of signature vectors' convex hull; remarkably, this simplex volume minimization approach has a solid theoretical ground for guaranteeing high-fidelity signature recovery under a very mild requirement on the data purity [25], [26].

An even faster fusion approach is based on deep learning (DL). Inspired by CNMF, the coupled unmixing network with a cross-attention mechanism (CuCaNet) [27] adopts the coupled convolutional autoencoder as the backbone, aiming at jointly decomposing the observed HSI/MSI into spectral bases and the corresponding coefficients. CuCaNet involves two main modules, i.e., cross-attention and spatial–spectral consistency; the former extracts spectral and spatial information from the observed HSI and MSI, respectively, for crossly sharing the information to both branches; the latter adopts two simple convolutional layers to simulate the HSI/MSI spatial blurring and spectral downsampling for adaptively learning spatial–spectral response functions for the subsequent fusion task. Another DL-based progressive zero-centric residual network (PZRes-Net) [28] was oriented from the wavelet analysis consisting of three main modules, including mean-invariant upsampling, progressive zero-centric residual learning, and refinement modules. Inspired by the HySure fusion theory, the deep blind iterative fusion network (DBIN) [29] estimates the spectral response transform and spatial spread transform using additional regularization to improve model generalization, where the forward model estimation and the HSI/MSI fusion are alternatively performed during the super-resolution reconstruction. As an interesting remark, the optimization-based CNMF and HySure have, respectively, inspired the DL-based CuCaNet and DBIN.

In fact, combining CO and DL has become a new trend in solving challenging imaging inverse problems, as reported in recent literature [30]. As a typical example, the MSI and HSI fusion network (MHF-Net) [31] was designed based on residual learning, the low-rank structure of the target image $\boldsymbol{Z}$, as well as the well-known proximal gradient optimization, to construct an interpretable network architecture. On the other hand, as simplex is a key topology in convex geometry, Dirichlet distribution with simplex domain naturally capturing the HSI data distribution [32] has been adopted in designing the unsupervised sparse Dirichlet network (USDN) [33] that is composed of two encoder–decoder networks, coupled by a shared decoder, for preserving the spectral information of HSI. As an extended version of USDN, the unregistered and unsupervised mutual Dirichlet network (U2MDN) [34] can fuse misaligned HSI/MSI using a single encoder–decoder structure for better robustness against the misalignment, and exploits the mutual information to prevent spectral distortion during the optimization procedure.

This category of methods has inspired our work, but we combine CO and DL in a radically new way as introduced in Section I-B.

B. Research Content, Paper Organization, and Notation

In this work, our fusion method is designed by employing both CO and deep neural network. Considering that remotely acquired HSI and MSI are often misaligned, the proposed deep network does not require the input HSI/MSI to be well aligned, making the challenging fusion network design even more difficult. In our proposed alignment-free HSI/MSI fusion method, we integrate a simple and efficient convex model to guide the fusion network, aiming to enhance the learning efficiency and performance, circumventing the need for the complex network structures typically found in conventional models. The

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convex model achieves fast learning of the spatial–spectral relation, and yields a rough estimate of the target image $\boldsymbol{Z}$. This allows the alignment-free fusion network to just focus on learning the difference between the rough solution and the target image $\boldsymbol{Z}$, similar to the idea of residual learning, greatly mitigating the burden of learning the spatial/spectral details from the misaligned HSI/MSI. The convex model, effectively guiding the **quasi-residual** learning task in our HSI/MSI fusion network, is a low-rank one well exploiting the convex geometry structure implicitly embedded in the hyperspectral signature space [24]. In the alignment-free fusion network, the group convolution is employed to more effectively incorporate the spectral relation between the observed HSI and MSI, and, in the meantime, to significantly reduce the number of network parameters for increasing the stability. Besides, by incorporating the proposed shifted window attention module (SWAM) with our fusion network, we accommodate both linear distortions, typical in cases of uniform shifts due to satellite orbital drift or Earth's rotation, and nonrigid distortions, arising from complex factors like satellite attitude changes or atmospheric variations. This versatility enhances the network's adaptability to real-world satellite imaging scenarios.

The remaining parts of this article are organized as follows. In Section II, we blend CO and DL to achieve alignment-free HSI/MSI fusion task, where the main idea is illustrated in Section II-A, followed by presenting the convex model and deep alignment-free fusion network in Sections II-B and II-C, respectively. In Section III, as verified by extensive experiments designed based on Wald's protocol [35], the proposed quasi-residual convex deep (QRCODE) network achieves the state-of-the-art alignment-free HSI/MSI fusion performance. Finally, concluding remarks are drawn Section IV.

Some standard notations used in this article are collectively defined here for better readability. $\mathbb{R}^{m\times n}$ is the set of all the real-valued $m \times n$ matrices. $\|\cdot\|_F$ and $\|\cdot\|_2$ stand for the Frobenius norm and ℓ_2-norm, respectively. Boldfaced $\mathbf{1}$ and $\mathbf{0}$ are, respectively, the all-one and all-zero vectors/matrices with proper dimensionality specified in the subscript. The notations "$\otimes$" and "$*$," respectively, stand for the Kronecker product and the convolution operator. For a given matrix $\boldsymbol{M} \in \mathbb{R}^{m\times n}$, its vectorized version is denoted by $\text{vec}(\boldsymbol{M}) \in \mathbb{R}^{mn}$ and is defined by the vector formed by sequentially stacking the columns of the matrix $\boldsymbol{M}$. Finally, $\boldsymbol{M}^T$ and $\boldsymbol{M}^{-1}$ denote the transpose and the inverse of the matrix $\boldsymbol{M}$, respectively.

II. Proposed Alignment-Free HSI/MSI Fusion Theory Blending Both CO and DL

In this section, we present the proposed fusion theory. In Section II-A, we illustrate the key idea behind the theory, which fuses the advantages from both CO and DL frameworks. We remark that blending the strengths of CO and DL is a new trend in solving challenging inverse imaging problems in the recent machine learning area; see [30] and the references therein. For the CO part, the learning criterion design and the associated algorithmic implementation are presented in Section II-B. For the DL part, the network design as well as the training details are presented in Section II-C.

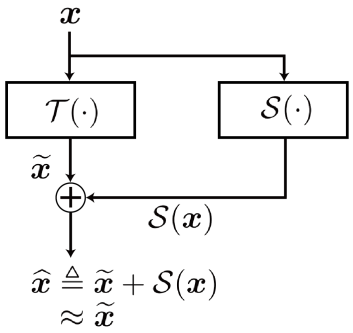

Fig. 1. Conceptual illustration of the proposed QRCODE for alignment-free HSI/MSI fusion. The convex model $\mathcal{T}(\cdot)$, often designed as a sophisticated deep network, is replaced by a convex model designed based on convex geometric structures embedded in the hyperspectral signature space, and is used to roughly estimate the target HSR HSI (denoted as $\widehat{\boldsymbol{x}}$) from the input data $\boldsymbol{x}$ (i.e., the misaligned HSI/MSI). The difference/residual between this rough estimate (denoted as $\widetilde{\boldsymbol{x}}$) and the target HSR HSI $\widehat{\boldsymbol{x}}$ is then learned by our alignment-free fusion network $\mathcal{S}(\cdot)$.

A. Key Idea Behind the Design of QRCODE

HSR image is critical for effective material classification, while HSI is essential for material identification. However, directly acquiring a HSR HSI is expensive. As the spatial resolution of MSI is often higher than HSI, fusing the HSR MSI with the LSR HSI becomes a popular alternative to obtain the desired HSR HSI. Nevertheless, directly designing a deep neural network to fuse HSI and MSI may not be effective, since the relation between the output target image (i.e., HSR HSI) and the input HSI/MSI may not be easily learned, especially when the two input images are not perfectly aligned; note that the misaligned issue greatly impacts on the fusion performance as reported in the remote sensing literature [3, Section 4.5].

From an intuitive ground, if we can roughly estimate the target image from the input HSI/MSI, then learning the relation between the output target image and the roughly estimated target image would be much easier. Such an intuition is also supported from the great success of residual learning [36], as learning the relation between the target image and its rough estimate can be thought of as the learning of the missing details—key philosophy behind the residual learning. The idea is graphically illustrated in Fig. 1, which shows a strategy similar, but not exactly the same, to the residual learning. This motivates us to break the fusion problem into two parts. Simply speaking, the first part is to compute a roughly estimated target image from the input HSI/MSI, while the second part is to learn the relation between the output target image and the roughly estimated target image. The first part and the second part will be achieved through CO and DL, respectively.

To be more specific, as remotely acquired HSI/MSI are often misaligned, the proposed deep neural network does not require the input HSI/MSI to be spatially co-registered. We need an effective convex model as guidance for HSI/MSI fusion learning tasks. To this end, we introduce a fast and simple convex model when learning the spatial–spectral relation for generating a rough solution. As alluded above, the rough solution may be regarded as a weak version of the target

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HSR HSI. Thus, the counterpart fusion network helps learn the missing details, i.e., the difference/residual between the target HSR HSI and the rough solution returned by the convex model. The convex model is designed to be a low-rank one for reducing the number of parameters, and well exploits the convex geometry structure implicitly embedded in the hyperspectral signature space, as will be detailed in Section II-B. The proposed QRCODE network is graphically illustrated in Fig. 2. The remaining parts of this section is organized as follows. In Section II-B, the convex model will be designed, and, in Section II-C, the alignment-free fusion network for the quasi-residual learning will be detailed.

B. Convex-Optimization-Enabled Model

Generally speaking, the model that provides knowledge/information is referred to as the prior knowledge for better learning efficiency of HSI/MSI fusion networks, while the model that exploits the knowledge/information during the learning phase is referred to as the alignment-free fusion network. Various applications include the semantic segmentation [37], face recognition [38], and object tracking [39], just to name a few. In this work, the aim of the convex model is to fast produce a roughly estimated target image (i.e., HSR HSI $\boldsymbol{Z}$), which provides crucial information of the target image $\boldsymbol{Z}$ that will be used to greatly mitigate the learning burden in the alignment-free fusion network, as can be seen from Fig. 2. In other words, with the information provided by the convex model, our HSI/MSI fusion model can simply focus on the learning of the residual (i.e., the difference between the target image $\boldsymbol{Z}$ and its rough estimate). This idea is somewhat similar to, but distinct from, the residual learning [36]. From the above illustration, one can see that the learning task of the convex model takes over the major responsibility, and hence the convex model should have stable performance, for which we will employ powerful first-order CO theory [22] as detailed below.

Let us denote the unknown target HSR HSI as $\boldsymbol{Z} \in \mathbb{R}^{M\times L}$, where M is the number of spectral bands and L is the number of pixels. Then, the observable LSR HSI can be modeled as $\boldsymbol{Y}_h = \boldsymbol{Z}\boldsymbol{B} \in \mathbb{R}^{M\times L_h}$, where $\boldsymbol{B}$ is the blurring matrix degrading the number of pixels to L_h [3]. Also, the observable HSR MSI can be modeled as $\boldsymbol{Y}_m = \boldsymbol{D}\boldsymbol{Z} \in \mathbb{R}^{M_m\times L}$, where $\boldsymbol{D}$ is the downsampling matrix degrading the number of spectral bands to M_m [3]. Thus, when $\boldsymbol{Y}_h$ and $\boldsymbol{Y}_m$ are perfectly aligned, a natural data-fitting criterion for recovering $\boldsymbol{Z}$ is to minimize $\|\boldsymbol{Y}_h - \boldsymbol{Z}\boldsymbol{B}\|_F^2 + \|\boldsymbol{Y}_m - \boldsymbol{D}\boldsymbol{Z}\|_F^2$.

Considering the misaligned issue, however, only one of the data-fitting term is valid. Without loss of generality, we will assume that $\boldsymbol{Z}$ and $\boldsymbol{Y}_m$ are spatially aligned, while $\boldsymbol{Z}$ and $\boldsymbol{Y}_h$ may not be perfectly aligned in both horizontal and vertical directions. In other words, the valid data-fitting term in the final convex criterion is $\|\boldsymbol{Y}_m - \boldsymbol{D}\boldsymbol{Z}\|_F^2$, which comes from the well-aligned observation. Before completing the convex criterion design, as well as the corresponding algorithm implementation, we first propose a simple and fast initialization strategy based on the affine set geometry [22]. Specifically, the target HSR HSI can be written as

$$\boldsymbol{Z} \approx \boldsymbol{C}\boldsymbol{S} + \boldsymbol{d}\boldsymbol{1}_L^T \tag{1}$$

for some coefficient matrix $\boldsymbol{S} \in \mathbb{R}^{b\times L}$, where $(\boldsymbol{C}, \boldsymbol{d}) \in \mathbb{R}^{M\times b} \times \mathbb{R}^M$ are, respectively, the basis matrix and data-mean vector computed from the observable HSI $\boldsymbol{Y}_h$. Note that it is reasonable to assume that the data/pixel affine hulls of both HSIs (i.e., $\boldsymbol{Z}$ and $\boldsymbol{Y}_h$) share almost the same affine basis $\boldsymbol{C}$, because all the pixels in both HSIs can be represented as convex combinations of same hyperspectral signature vectors [24, Section 2]; similarly, from the blurring model $\boldsymbol{Y}_h = \boldsymbol{Z}\boldsymbol{B}$, it is reasonable to assume that both HSIs also share almost the same data-mean vector $\boldsymbol{d}$ [3]. Though using the affine basis and data-mean vector of $\boldsymbol{Y}_h$ to approximate those of $\boldsymbol{Z}$ may lead to some precision error, this is a minor issue as this is just for the initialization of the iterating algorithm to be developed below, meaning that the error would be diminished during the algorithmic iterations. Now, from $\boldsymbol{Y}_m = \boldsymbol{D}\boldsymbol{Z}$, we have

$$\boldsymbol{Y}_m - \boldsymbol{D}\boldsymbol{d}\boldsymbol{1}_L^T \approx \boldsymbol{D}\boldsymbol{C}\boldsymbol{S}. \tag{2}$$

We purposely choose the number of columns in the basis matrix $\boldsymbol{C}$ as $b = M_m$, which retains 99% signal energy in the eigenimages (i.e., rows of $\boldsymbol{S}$), because a higher value leads to an underdetermined system, while a lower value may waste some information embedded in $\boldsymbol{Y}_m$. Another advantage behind such a choice is that $\boldsymbol{D}\boldsymbol{C}$ will be a square matrix, so that one can easily estimate

$$\widehat{\boldsymbol{S}} = (\boldsymbol{D}\boldsymbol{C})^{-1}(\boldsymbol{Y}_m - \boldsymbol{D}\boldsymbol{d}\boldsymbol{1}_L^T) \tag{3}$$

thereby leading to $\widehat{\boldsymbol{Z}} = \boldsymbol{C}\widehat{\boldsymbol{S}} + \boldsymbol{d}\boldsymbol{1}_L^T$ that will be used to initialize the algorithm to be developed below.

Recall that due to the misaligned issue, the aforementioned data-fitting criterion is

$$\min_{\boldsymbol{Z}} \ \|\boldsymbol{Y}_m - \boldsymbol{D}\boldsymbol{Z}\|_F^2. \tag{4}$$

As for the other traditional data-fitting term corresponding to the observation $\boldsymbol{Y}_h$, it contributes to the computation of the basis $\boldsymbol{C}$ as discussed above, and hence helps reduce the number of variables (i.e., from $\boldsymbol{Z}$ to $\boldsymbol{S}$) for mitigating the ill-posed nature of the fusion problem as detailed below. All in all, we arrive at the following convex fusion criterion:

$$\min_{\boldsymbol{Z}} \ \|\boldsymbol{Y}_m - \boldsymbol{D}\boldsymbol{Z}\|_F^2 + \Pi(\boldsymbol{Z}) \tag{5}$$

where Π is the indicator function corresponding to the box constraint for forcing the entries of $\boldsymbol{Z}$ to match the nature of the reflectance lying within the interval $[0, 1]$, i.e., $\Pi(\boldsymbol{Z}) = 0, \ \forall \boldsymbol{Z} \in [0, 1]^{M\times L}$ (∞, otherwise). However, as alluded before, there are too many variables in the target image $\boldsymbol{Z}$. Just like the success of convolutional neural network (CNN) lying within the greatly reduced number of network parameters, we first reformulate the above convex fusion criterion in terms of $\boldsymbol{S}$, thereby allowing us to reduce the number of variables from ML to M_mL; note that in practice, the number of multispectral bands M_m is much smaller than the number of hyperspectral bands M. Besides reducing the number of parameters, our reformulation will be based on the CO theory, known as ADMMs in statistical machine learning [23]. Specifically, the above fusion criterion is reformulated into the following ADMM-form [22], i.e.,

$$\min_{\boldsymbol{T}=\boldsymbol{C}\boldsymbol{S}} \ \|\boldsymbol{\Phi} - \boldsymbol{\Psi}\boldsymbol{S}\|_F^2 + \widetilde{\Pi}(\boldsymbol{T}) \tag{6}$$

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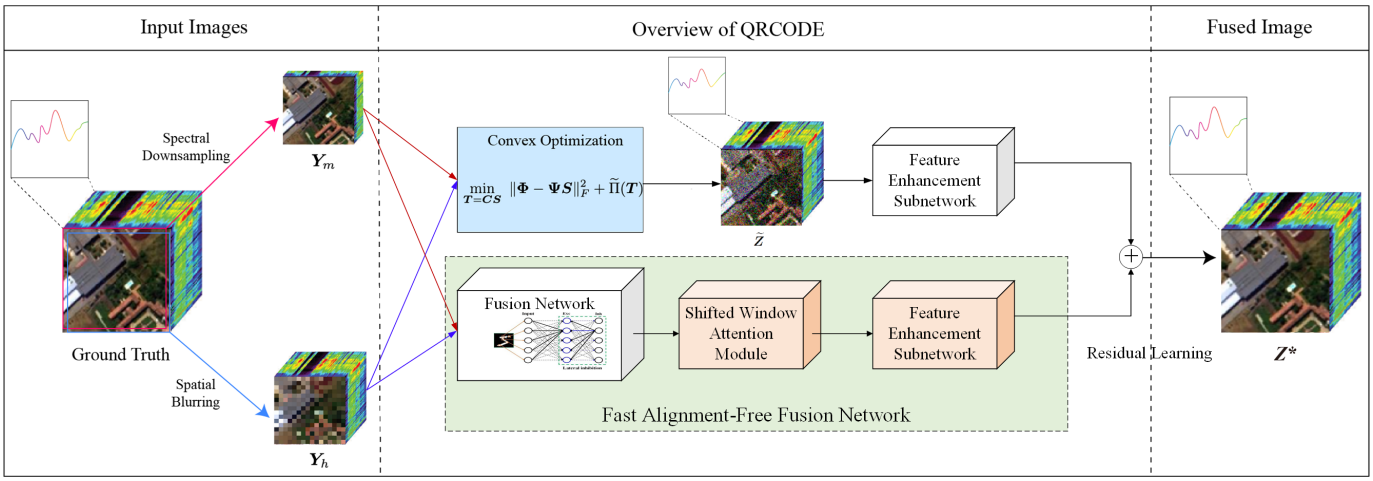

Fig. 2. Overview of the proposed QRCODE. The misaligned images, i.e., the HSR MSI $\boldsymbol{Y}_m$ and the LSR HSI $\boldsymbol{Y}_h$, correspond to different spatial regions. Both images are fed into the convex model, followed by computing the rough estimate of the target HSR HSI $\boldsymbol{Z}$ using CO algorithm, i.e., Algorithm 1. The resulting rough estimate $\widetilde{\boldsymbol{Z}}$ provides crucial information of $\boldsymbol{Z}$ for the alignment-free fusion network to effectively conduct the quasi-residual learning, i.e., the learning of the difference between the target image $\boldsymbol{Z}$ and its rough estimate $\widetilde{\boldsymbol{Z}}$.

where $\boldsymbol{\Phi} \triangleq \boldsymbol{Y}_m - \boldsymbol{D}\boldsymbol{d}\boldsymbol{1}_L^T$, $\boldsymbol{\Psi} \triangleq \boldsymbol{D}\boldsymbol{C}$, and $\widetilde{\Pi}$ is the indicator function of the shifted box constraint set "$[0,1]^{M\times L} - \boldsymbol{d}\boldsymbol{1}_L^T$." Then, according to the ADMM theory [22], [23], the CO algorithm for solving the convex criterion (1) is given in Algorithm 1, where $\mu > 0$ and $\boldsymbol{V}$ denote the penalty parameter and scaled dual variable, respectively.

Algorithm 1 Convex Model for Solving (6)

1: **Given** $\mu > 0$. Initialize $\boldsymbol{S}_0$ via (3), $\boldsymbol{V}_0 := \boldsymbol{0}_{M\times L}$, and $q := 0$.
2: **repeat**
3: $\boldsymbol{T}_{q+1} := \arg\min_{\boldsymbol{T}} \ \widetilde{\Pi}(\boldsymbol{T}) + \frac{\mu}{2}\|\boldsymbol{C}\boldsymbol{S}_q - \boldsymbol{T} - \boldsymbol{V}_q\|_F^2$;
4: $\boldsymbol{S}_{q+1} := \arg\min_{\boldsymbol{S}} \ \|\boldsymbol{\Phi} - \boldsymbol{\Psi}\boldsymbol{S}\|_F^2 + \frac{\mu}{2}\|\boldsymbol{C}\boldsymbol{S} - \boldsymbol{T}_{q+1} - \boldsymbol{V}_q\|_F^2$;
5: $\boldsymbol{V}_{q+1} := \boldsymbol{V}_q - (\boldsymbol{C}\boldsymbol{S}_{q+1} - \boldsymbol{T}_{q+1})$;
6: $q := q + 1$;
7: **until** the predefined stopping criterion is met.
8: **Output** $\boldsymbol{S}^\star := \boldsymbol{S}_q$, as well as the rough estimate of the target HSI $\widetilde{\boldsymbol{Z}} := \boldsymbol{C}\boldsymbol{S}^\star + \boldsymbol{d}\boldsymbol{1}_L^T$ (see (1)).

To apply Algorithm 1 to efficiently obtain the rough estimate of the target HSI as $\widetilde{\boldsymbol{Z}} := \boldsymbol{C}\boldsymbol{S}^\star + \boldsymbol{d}\boldsymbol{1}_L^T$, we still need to derive closed-from solutions of $(\boldsymbol{T}_{q+1}, \boldsymbol{S}_{q+1})$ as follows. For $\boldsymbol{T}_{q+1}$, it is nothing but the projection of $\boldsymbol{C}\boldsymbol{S}_q - \boldsymbol{V}_q$ onto the shifted box "$[0,1]^{M\times L} - \boldsymbol{d}\boldsymbol{1}_L^T$." Such projection can be performed column by column; specifically, by letting $\boldsymbol{t}_j$ (resp., $\boldsymbol{\delta}_j$) be the jth column of $\boldsymbol{T}_{q+1}$ (resp., $\boldsymbol{C}\boldsymbol{S}_q - \boldsymbol{V}_q$), we have

$$\boldsymbol{t}_j := [\mathrm{Proj}_{[0,1]^M}(\boldsymbol{\delta}_j + \boldsymbol{d})] - \boldsymbol{d} \quad \forall j = 1, \ldots, L \tag{7}$$

where $\mathrm{Proj}_{[0,1]^M}$ projects the input vector onto the box $[0,1]^M$. Then, $\boldsymbol{T}_{q+1}$ can be updated as

$$\boldsymbol{T}_{q+1} := [\boldsymbol{t}_1, \ldots, \boldsymbol{t}_L]. \tag{8}$$

As for the update of $\boldsymbol{S}_{q+1}$, we equivalently derive its vectorized version for convenience

$$\begin{aligned}
&\mathrm{vec}(\boldsymbol{S}_{q+1}) \\
&\quad \equiv \arg\min_{\boldsymbol{s}} \ \|\boldsymbol{\phi} - (\boldsymbol{I}_L \otimes \boldsymbol{\Psi})\boldsymbol{s}\|_2^2 + \frac{\mu}{2}\|(\boldsymbol{I}_L \otimes \boldsymbol{C})\boldsymbol{s} - \boldsymbol{\theta}\|_2^2 \\
&\quad = [\boldsymbol{I}_L \otimes \boldsymbol{\Gamma}]^{-1}\left[(\boldsymbol{I}_L \otimes \boldsymbol{\Psi}^T)\boldsymbol{\phi} + \frac{\mu}{2}(\boldsymbol{I}_L \otimes \boldsymbol{C}^T)\boldsymbol{\theta}\right] \\
&\quad = \left[\boldsymbol{I}_L \otimes \left(\boldsymbol{\Psi}^T\boldsymbol{\Psi} + \frac{\mu}{2}\boldsymbol{I}_{M_m}\right)^{-1}\right]\mathrm{vec}\left(\boldsymbol{\Psi}^T\boldsymbol{\Phi} + \frac{\mu}{2}\boldsymbol{C}^T\boldsymbol{\Theta}\right) \\
&\quad = \mathrm{vec}\left(\left(\boldsymbol{\Psi}^T\boldsymbol{\Psi} + \frac{\mu}{2}\boldsymbol{I}_{M_m}\right)^{-1}\left(\boldsymbol{\Psi}^T\boldsymbol{\Phi} + \frac{\mu}{2}\boldsymbol{C}^T\boldsymbol{\Theta}\right)\right)
\end{aligned} \tag{9}$$

where $\boldsymbol{\phi} \triangleq \mathrm{vec}(\boldsymbol{\Phi})$, $\boldsymbol{\Gamma} \triangleq \left(\boldsymbol{\Psi}^T\boldsymbol{\Psi} + (\mu/2)\boldsymbol{C}^T\boldsymbol{C}\right)$, $\boldsymbol{\Theta} \triangleq \boldsymbol{T}_{q+1} + \boldsymbol{V}_q$, and $\boldsymbol{\theta} \triangleq \mathrm{vec}(\boldsymbol{\Theta}) = \mathrm{vec}(\boldsymbol{T}_{q+1} + \boldsymbol{V}_q)$. Then, one can update $\boldsymbol{S}_{q+1}$ using

$$\boldsymbol{S}_{q+1} := \left(\boldsymbol{\Psi}^T\boldsymbol{\Psi} + \frac{\mu}{2}\boldsymbol{I}_{M_m}\right)^{-1}\left(\boldsymbol{\Psi}^T\boldsymbol{\Phi} + \frac{\mu}{2}\boldsymbol{C}^T\boldsymbol{\Theta}\right). \tag{10}$$

Therefore, the design of the convex model, as well as its implementation, has been completed.

As aforementioned, the convex model just aims at returning a rough estimate of the target HSR HSI $\boldsymbol{Z}$, for guiding the learning of the residual (i.e., the difference between the target image and its rough estimate) in the alignment-free fusion network; see Fig. 2. So, the convex model does not need to sophisticatedly return a precisely estimated target image $\boldsymbol{Z}$, meaning that though there may have space to further upgrade the fusion performance of the convex model, it is not the key toward the success of the proposed QRCODE. In Section II-C, we detail the design of the alignment-free fusion network.

C. Proposed FAFNet

This section presents the benefits and technical aspects of the proposed fast alignment-free fusion network (FAFNet) in QRCODE for alignment-free HSI/MSI fusion. The comprehensive network architecture of the suggested alignment-free

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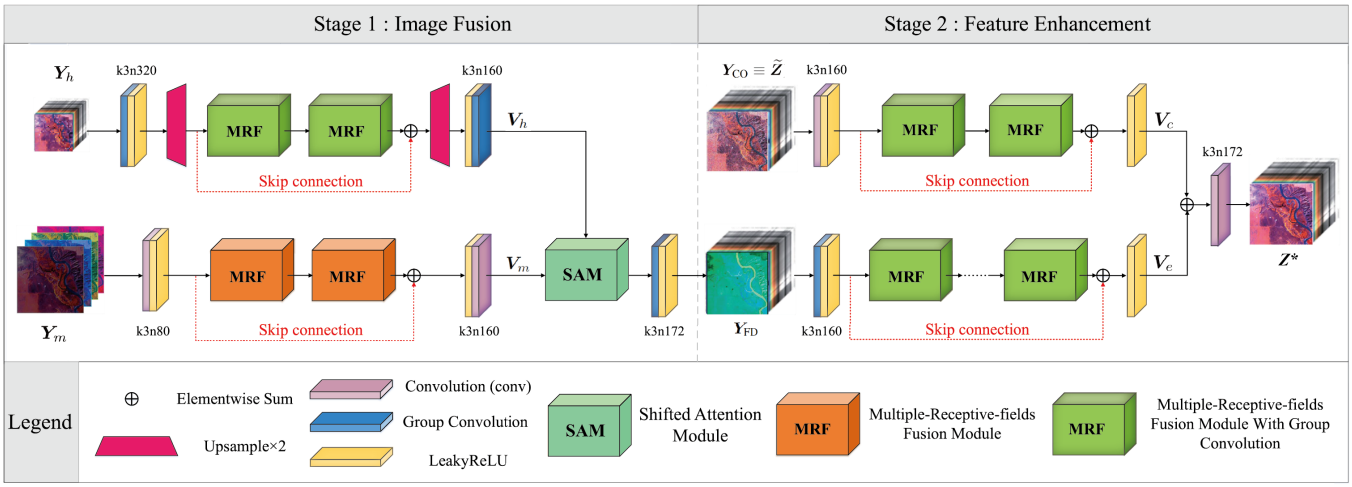

Fig. 3. Network architecture of the proposed FAFNet in QRCODE, where "k(s)n(c)" means that the corresponding convolution layer involves c channels with a kernel size of s, and $\widetilde{\boldsymbol{Z}}$ is returned from the convex model (see Algorithm 1).

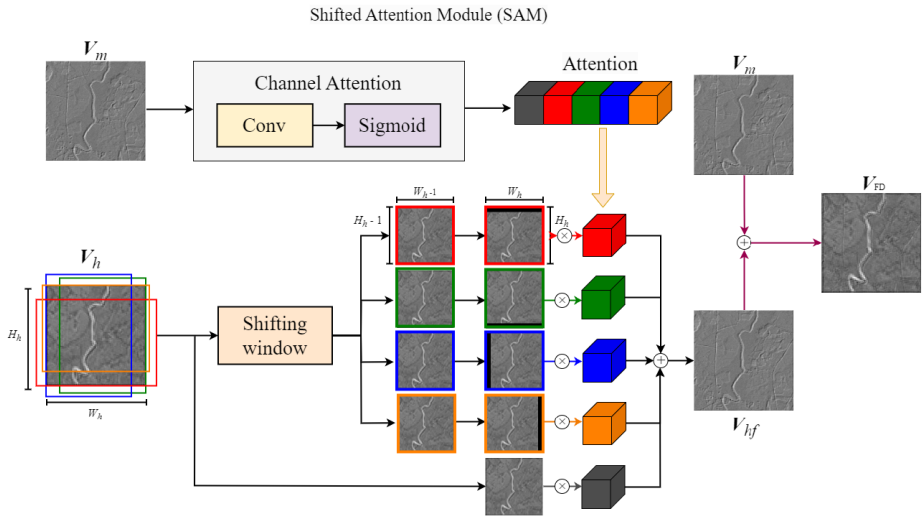

Fig. 4. Proposed SWAM for alignment-free hyperspectral and MSI fusion by exploiting neighboring feature embedding learning.

student network within our QRCODE is depicted in Fig. 3. Initially, the LSR HSI $\boldsymbol{Y}_h$ and HSR MSI $\boldsymbol{Y}_m$ are input into their respective subnetworks based on the proposed multiple-receptive-fields fusion (MRF) module, which simultaneously enhances their spatial–spectral resolution. The core of misalignment fusion within our network is the proposed SWAM, which intelligently uncovers the spatially neighboring correlation through a learnable attention mechanism to maintain high-performance results as the misalignment degree rises. By investigating the co-redundancy of spectral correspondence between LSR HSI and HSR MSI, we introduce grouped convolution in our QRCODE to decrease the number of parameters without compromising performance. Subsequently, the feature responses of the LSR HSI and HSR MSI are combined as the network's first-stage output, referred to as the fused draft (FD) $\boldsymbol{Y}_{\mathrm{FD}}$. To attain excellent subjective and objective quality of the reconstructed HSR HSI $\boldsymbol{Z}^*$, a high-performance and complex network is typically required for recovering the HSR HSI from FD. Motivated by the conventional teacher–student (TS) model, but rather than adopt sophisticated networks in teacher models, we incorporate the HSR HSI $\boldsymbol{Y}_{\mathrm{CO}}$ produced by our simple convex model (i.e., $\boldsymbol{Y}_{\mathrm{CO}} \equiv \widetilde{\boldsymbol{Z}}$) with a relatively simple subnetwork to guide the feature learning of the subnetwork for $\boldsymbol{Y}_{\mathrm{FD}}$. This significantly reduces the number of convolutional layers while maintaining good performance. This approach is equivalent to extracting the feature representation from $\boldsymbol{Y}_{\mathrm{CO}}$, facilitating the minimized complexity of the proposed QRCODE design. The specifics of the proposed FAFNet in our QRCODE are provided below.

1) Shifted Window Attention Module: In this work, the proposed SWAM is demonstrated to effectively address the fusion of feature maps of HSR MSI and LSR HSI, even in the absence of strict coregistration. Let the feature representation of HSR MSI and LSR HSI be denoted as $\boldsymbol{V}_m$ and $\boldsymbol{V}_h$, respectively. When these feature maps lack strict coregistration, direct fusion through element-wise addition $\boldsymbol{V}_{\mathrm{FD}} = \boldsymbol{V}_m + \boldsymbol{V}_h$ may be susceptible to misalignment. To tackle this issue, the SWAM has been designed, as depicted in Fig. 4. Given the features of HSR MSI and LSR HSI, $\boldsymbol{V}_m$ and $\boldsymbol{V}_h$, $\boldsymbol{V}_h$ is cropped into four subfeatures of size $H_h - 1 \times W_h$ or $H_h \times W_h - 1$ using a shifting window (SW). More specifically, the SW operation

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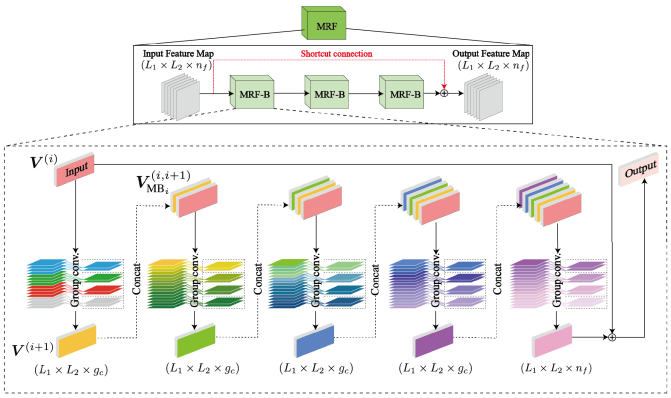

Fig. 5. Detailed network architecture of the proposed lightweight MRF with grouped convolution.

can be expressed as

$$\boldsymbol{V}_{hs}(i,j) = \boldsymbol{V}_h(i+a, j+b) \tag{11}$$

where $a \in 0,1$, $b \in 0,1$, and $i = 0,1,\ldots,H_h - a$ and $j = 0,1,\ldots,W_h - b$. This equation represents the SW-based cropping operation incorporating a 1-pixel shift along the height or width axes. For example, a cropped feature $\boldsymbol{V}_{hs}$, shifted by 1 pixel to the left, can be obtained by setting $a = 0$ and $b = 1$. This method enables the utilization of neighboring feature embeddings for $\boldsymbol{V}_h$ to assist the fusion task in the presence of misalignment.

Subsequently, zero-padding is applied to reshape each sub-feature to $H_h \times W_h$. Evaluating the importance of each subfeature for fusion is essential. For this purpose, the channel attention derived from $\boldsymbol{V}_m$ is proposed to intelligently learn the significance of each sub-feature in a data-driven manner. Specifically, the element-wise attention $\mathbf{A} \in \mathbb{R}^{5\times H_h \times W_h}$ is defined as

$$\mathbf{A} = \sigma(\text{Conv.}(\boldsymbol{V}_m)) \tag{12}$$

where Conv. denotes the convolution operation and σ represents the sigmoid activation. The initial convolution is utilized to project the channels of $\boldsymbol{V}_m$ onto a smaller dimension, effectively reducing computational complexity. In this way, each element of $\mathbf{A}$ possesses an attention value, enabling the proposed SWAM to learn the importance of each subfeature for enhanced performance.

Finally, the padded subfeature of $\boldsymbol{V}_h$ is re-weighted to obtain the weighted neighboring feature embedding

$$\boldsymbol{V}_{hf} = \boldsymbol{V}_h \times \mathbf{A}^{(0)} + \sum_{i=1}^{4} \boldsymbol{V}_{hs}^{(i)} \times \mathbf{A}^{(i)}. \tag{13}$$

Subsequently, the FD feature $\boldsymbol{V}_{\text{FD}} = \boldsymbol{V}_m + \boldsymbol{V}_{hf}$ is acquired. This approach allows for the intelligent exploitation of neighboring features when misalignment arises, thus improving performance and stability.

2) Lightweight Design of Our FAFNet: In contrast to traditional TS models, the convex model in our proposed QRCODE significantly reduces the computational complexity of the convex subnetwork while maintaining the advantages of our convex-optimization-aware solution $\boldsymbol{Y}_{\text{CO}} \equiv \widetilde{\boldsymbol{Z}}$ (see Algorithm 1). First, the FD $\boldsymbol{Y}_{\text{FD}}$ generated by our SWAM is enhanced by several convolutional blocks to form the fused feature $\boldsymbol{V}_e$ (see Stage 2 in Fig. 3), and concurrently, the $\boldsymbol{Y}_{\text{CO}}$ is transformed into the same feature domain $\boldsymbol{V}_c$ using a compact subnetwork followed after the convex model (see Stage 2 in Fig. 3). On the one hand, due to the significantly better spectral resolution than spatial resolution in $\widetilde{\boldsymbol{Z}}$, the proposed FAFNet can be effectively guided by $\boldsymbol{V}_c$ to generate high-fidelity spectral signals. On the other hand, learning to recover the residuals between the target image $\boldsymbol{Z}$ and its rough estimate $\widetilde{\boldsymbol{Z}}$ in the spatial domain is relatively simple [36] compared to directly learning $\boldsymbol{Z}$ from $\boldsymbol{Y}_{\text{FD}}$.

A practical and powerful base block for constructing our FAFNet is essential. Conventional residual block (ResBlock) may be ineffective for image super-resolution tasks [40]; thus, various base blocks have been proposed to enhance the performance of image restoration fields. Inspired by [41], we introduce the powerful MRF module with grouped convolution to fully explore the multiscale feature representations, as shown in Fig. 5. Nonetheless, the naive design of MFA is detrimental to lightweight purposes. We describe it below. Each MRF consists of three multi-receptive-field blocks (MRF-B) in the naive design in [41] with two residual connections. However, overusing the residual connection may not significantly improve performance but cause higher memory usage. Therefore, we remove one of the residual connections from MRF to reduce memory usage. In each MRF-B, the incoming feature map $\boldsymbol{V}^{(i)} \in \mathbb{R}^{L_1\times L_2\times n_f}$ is projected into a squeezed one $\boldsymbol{V}^{(i+1)}$ with g_c channels, followed by concatenating (denoted by "cat") $\boldsymbol{V}^{(i)}$ and $\boldsymbol{V}^{(i+1)}$ along the channel dimension to obtain the aggregated feature map $\boldsymbol{V}_{\text{MB}_i}^{(i,i+1)} = \text{cat}(\boldsymbol{V}^{(i)}, \boldsymbol{V}^{(i+1)})$. This design enables diverse receptive fields to be explored in a single MRF, resulting in more effective and efficient feature representation learning. Note that the last grouped convolution will yield a $L_1 \times L_2 \times n_f$ output feature map.

Second, we argue that the full convolution operation may be unnecessary for hyperspectral fusion tasks. The convolution operation is defined as

$$f * g(i,j) = \sum_{m,n} f(i+m, j+n) \times g(m,n) \tag{14}$$

where $m = [-(k-1)/2, (k-1)/2]$ and $n = [-(k-1)/2, (k-1)/2]$ denote the kernel size k indicators, g represents the kernel function, and f refers to the input data. Typically, a $k \times k \times c_{\text{in}} \times c_{\text{out}}$ kernel generates the c_{out} output feature maps from the c_{in} input feature maps, implying that all input feature map channels contribute to an output feature map.

However, fusing HSR MSI and LSR HSI in the feature domain remains redundant in the spectral signature space. Specifically, only a few LSR HSI bands strongly correlate with an HSR MSI band. For instance, as illustrated in the input branch in Fig. 5, the first HSR MSI band (indicated in blue) strongly correlates with a few LSR HSI bands. Since the full convolution operation (14) generates a feature map from all LSR HSI bands, the less-relevant LSR HSI bands contribute minimally, suggesting that the convolution operation for these irrelevant bands can be eliminated to enhance efficiency. This

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insight motivates us to develop a grouped-convolution-aware network architecture [42] for LSR HSI to reduce computational complexity without sacrificing performance.

Employing grouped convolution, the input and output feature maps are divided into c_g groups, followed by performing the convolution operation group-wise. Specifically, for c_g kernel groups, with each kernel $g \in \mathbb{R}^{k\times k\times(c_{\text{in}}/c_g)\times(c_{\text{out}}/c_g)}$, the kth output feature map group is calculated solely from the kth input feature map group. It is important to note that $c_g = 4$ in this study. Consequently, by leveraging the redundancy between HSI/MSI for our FAFNet, we can reduce computational complexity without compromising performance, resulting in a cost-effective and efficient approach to generate HSR HSI using our QRCODE. From a technical standpoint, only the transitional layer, SWAM and MRF for HSR MSI employ full convolution, while the remaining components utilize grouped convolution. In the MRF for HSR MSI, the number of bands is relatively limited, leading to less redundancy between the spectrum, and thus, full convolution may be a more suitable option.

Finally, the reconstructed HSR HSI $\boldsymbol{Z}^*$ generated by our QRCODE can be obtained by

$$\boldsymbol{Z}^* = \text{Conv}(\boldsymbol{V}_c + \boldsymbol{V}_e) \tag{15}$$

where Conv aims to generate the target HSR HSI from the feature representations of the convex-based solution and our FAFNet's result. The loss function is defined by the $\ell 1$ distance and regularizer as follows:

$$J(\boldsymbol{Z}, \boldsymbol{Z}^*) = ||\boldsymbol{Z} - \boldsymbol{Z}^*||_1 + \alpha \times \text{std}(\boldsymbol{Z}^*) \tag{16}$$

where std denotes the standard deviation of the reconstructed HSR HSI $\boldsymbol{Z}^*$, and the parameter α controls the balance between the data and the regularizer terms. In this study, α is set to $1e^{-5}$. This regularizer can be regarded as the weight decay policy to mitigate overfitting during our QRCODE and enhance stability.

III. Experimental Results

A. Settings

We first describe the dataset preparation and how the LSR HSI and HSR MSI were synthesized in the experiments. The dataset used for performance evaluation was acquired by airborne visible/infrared imaging spectrometer (AVIRIS) sensor [43]. The collected dataset includes various natural landscapes from the US/Canada, such as cities, mountains, lakes, fields and plants, which were captured from 2006 to 2011. The original HSI image is partitioned into non-overlapping 259×259 HSI sub-images with 224 bands covering a wavelength range from 400 to 2500 nm. The low-quality bands (1–10, 104–116, 152–170, and 215–224) are removed as suggested in [41]. Therefore, the number of the bands of HSI images is 172. Generating the HSR MSI and LSR HSI from HSR HSI is also essential. Wald's protocol [35] is used to simulate the image fusion experiments, where the 256×256 HSR HSI $\boldsymbol{Y}$ is cropped from the HSI sub-image starting at the top-left corner. Two different downsampling matrices are used to synthesize MSI images $\boldsymbol{Y}_{m4} \in \mathbb{R}^{256\times256\times4}$ and $\boldsymbol{Y}_{m6} \in \mathbb{R}^{256\times256\times6}$ with four and six multispectral bands, respectively. One downsampling matrix $\boldsymbol{D}_4 \in \mathbb{R}^{4\times172}$ is approximately corresponding to Landsat TM bands 1–4 (covering 450–520, 520–600, 630–690, and 770–900 nm). The other downsampling matrix, $\boldsymbol{D}_6 \in \mathbb{R}^{6\times172}$, roughly corresponds to Landsat TM bands 1–5 and 7 (covering 450–520, 520–600, 630–690, 770–900, 1550–1750, and 2090–2350 nm).

In addressing image misalignment challenges in HSI/MSI fusion, our study integrates linear and nonrigid simulation methods to represent real-world scenarios effectively, as follows.

1) *Linear Simulation in Satellite Imagery Analysis:* We utilize linear simulation to model uniform shifts in satellite imagery, typically caused by orbital drift, Earth's rotation, or minor positional changes during capture.
2) *Nonrigid Simulation for Complex Deformations:* For more intricate nonlinear distortions resulting from satellite attitude changes or atmospheric disturbances, our research applies nonrigid simulation methods, modeled using a bimodal Gaussian approach (see Section III-D).

These integrated simulation methods provide a nuanced analysis of satellite image misalignments, enhancing the accuracy and applicability of our HSI/MSI fusion model in diverse satellite imaging scenarios.

The Gaussian point spread function with variance $\sigma = 3$ and blurring factor $b_r = 4$ is used to generate a spatially degenerate matrix $\boldsymbol{B} \in \mathbb{R}^{L^2\times L_l^2}$ to synthesize the aligned and misaligned LSR HSIs. The aligned LSR HSI $\boldsymbol{Y}_{h1} \in \mathbb{R}^{64\times64\times172}$ is obtained by $\boldsymbol{Y}$ with the spatially degenerate matrix $\boldsymbol{B}$. Likewise, the misaligned LSR HSI images could be cropped from the original HSI sub-image at the starting point (o_x, o_y). Specifically, the $\boldsymbol{Y}_{h2}$, $\boldsymbol{Y}_{h3}$, and $\boldsymbol{Y}_{h4}$ represent the 1-, 2-, and 3-pixel misaligned LSR HSI images, respectively. Finally, HSR MSI and LSR HSI images are contaminated by additive Gaussian noise with zero mean noise with signal-to-noise ratios of 30 and 35 dB, respectively [17].

The proposed method is implemented on Mathworks MATLAB R2020b and PyTorch for the convex model and our FAFNet. All the experiments are conducted on a personal computer with an Intel i9-10900K CPU, 128 GB of system memory, and an NVIDIA GTX-3090 GPU. The learning rate and the number of the total epochs are $1e-3$ and 600. The step learning scheduling reduces the learning rate by a decay factor of 10 every 150 epochs. The ℓ_1-norm function is adopted in the training phase with Adam optimizer [44]. To learn the misalignment property for SWAM, we randomly shift the LSR MSI by 1 pixel in either X- or Y-axis, termed mixed-training. The number of the collected HSR HSI is 2078. We randomly partition the collected dataset into training, validation, and testing sets for performance evaluation, where the number of the training, validation, and testing samples are 1678, 200, and 200, respectively. The spatial/spectral resolutions of HSR MSI and LSR HSI are $256 \times 256 \times M_m$ and $64 \times 64 \times 172$, where M_m is either 4 or 6 in our experiments. Note that the number of channels of the input and output feature maps should be divisible by c_g for our QRCODE; otherwise, the full convolution will be used instead.

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TABLE I
MODIFIED HYPERPARAMETERS FOR OUR HSI/MSI FUSION TASK

Method	Implementation Details and Hyper-Parameter Setting
PZRes-Net	learning rate $= 0.5 \times 10^{-3}$ blurring factor: 4 Stage1: cin=the number of MSI bands, cout=64 Stage2: cin=64+the number of MSI bands, cout=128 Stage3: cin=128+the number of MSI bands, cout=172
DBIN	blurring factor: 4 $0 \leq$iter$\leq 2 \times 10^4$, learning rate $= 4 \times 10^{-}5$ $2 \times 10^4 <$iter$\leq 6 \times 10^4$, learning rate $= 2 \times 10^{-}5$ $6 \times 10^4 <$iter$\leq 1.4 \times 10^5$, learning rate $= 1 \times 10^{-}5$ $1.4 \times 10^5 <$iter$\leq 2.5 \times 10^5$, learning rate $= 5 \times 10^{-}6$ Convolution and Deconvolution(kernel size: 6, stride: 4)

For the computational and space complexity comparison, the number of parameters (#Param.), and the floating-point operations (FLOPs) are used to evaluate the efficiency of the deep-learning-based schemes. Meanwhile, peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), *erreur relative globale adimensionnelle de synthése* (ERGAS), and root mean-squared error (RMSE) are adopted in our experiments for performance evaluation. Note that $\boldsymbol{Y}$ is the reference image, and $\boldsymbol{Z}$ is the obtained fused HSR HSI. The definition of each metric is listed as follows.

1) PSNR is defined as

$$\text{PSNR} = \frac{1}{M}\sum_{m=1}^{M}\text{PSNR}_m. \tag{17}$$

Note that PSNR_m measures the spatial quality of a single band and m represents the mth band

$$\text{PSNR}_m = 10\log_{10}\left(\frac{\max\{z_{mn}^2 \mid n \in \mathcal{I}_L\}}{\frac{1}{L}\|\boldsymbol{Z}^{(m)} - \boldsymbol{Y}^{(m)}\|_2^2}\right) \tag{18}$$

where z_{mn} denotes the nth entry in the vector $\boldsymbol{Z}^{(m)}$, and $\mathcal{I}_L \triangleq \{1, \ldots, L\}$. A higher PSNR value indicates a better spatial quality of the fused image $\boldsymbol{Z}$.

2) SAM is defined as

$$\text{SAM} = \frac{1}{L}\sum_{n=1}^{L}\arccos\left(\frac{(z[n])^T y[n]}{\|z[n]\|_2 \cdot \|y[n]\|_2}\right) \tag{19}$$

where $z[n]$ denotes the nth column of $\boldsymbol{Z}$. The smaller the absolute value of SAM, the better the spectral reconstruction of $\boldsymbol{Z}$.

3) RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{M}\sum_{m=1}^{M}\text{RMSE}_m^2} \tag{20}$$

where

$$\text{RMSE}_m = \frac{1}{\sqrt{L}}\|\boldsymbol{Z}^{(m)} - \boldsymbol{Y}^{(m)}\|_2. \tag{21}$$

The smaller the RMSE value, the better the global quality of the fused image $\boldsymbol{Z}$.

4) ERGAS is defined as

$$\text{ERGAS} = \frac{100}{r}\sqrt{\frac{1}{M}\sum_{m=1}^{M}\frac{\text{RMSE}_m^2}{\mu_{\boldsymbol{Y}^{(m)}}^2}} \tag{22}$$

where r is the resolution ratio of the LSR HSI and the HSR MSI and $\mu_{\boldsymbol{Y}}^{(m)}$ is the mean of $\boldsymbol{Y}^{(m)}$. The smaller the ERGAS, the smaller the spectral distortion.

B. Performance Evaluation

In our experiments, we consider nine cutting-edge HSI/MSI fusion schemes, including MHF-Net [31], PZRes-Net [28], CUCaNet [27], USDN [33], DBIN [29], U2MDN [34], HySure [19], BIM [17], and CO-CNMF [3]. It is noteworthy that the final three approaches are optimization-based schemes. All data-driven methods are trained using identical training settings (i.e., training data and pre-processing strategy). As the data utilized in PZRes-Net [28] and DBIN [29] differ from ours, we modify their hyperparameters to fit our tasks, with the primary adjustments summarized in Table I. The performance of PZRes-Net [28] and DBIN [29] under the modified settings exceeds that of their default settings. It should be noted that the results reported in this article are marked in red and blue to indicate the best and the runner-up performance indices, respectively.

Tables II and III present the objective quality assessment between the proposed QRCODE and other state-of-the-art techniques in terms of PSNR, SAM, RMSE, and ERGAS. The QRCODE significantly outperforms other advanced methods, particularly in misalignment scenarios, due to the neighboring feature embedding of the proposed SWAM. Even in aligned situations, our QRCODE exhibits promising results compared to DBIN [29], especially regarding the SAM metric. The substantial advantages of the convex model for spectral information reconstruction in fusion tasks are also validated. Furthermore, the lightweight design of QRCODE enhances efficiency, attributable to the stability offered by the convexity of the convex model. The QRCODE also considerably surpasses DBIN [29] as misaligned pixels increase. Although BIM [17] demonstrates good performance in misalignment cases, extended runtimes (e.g., exceeding 80 s) pose challenges for real-time applications. By exploiting alignment-free properties and convex solution guidance, the QRCODE achieves an optimal balance between performance and efficiency for both aligned and misaligned cases.

The comparison of visual quality for reconstructed HSR HSI using the QRCODE and alternative fusion techniques in the presence of a 3-pixel misalignment in LSR HSI is illustrated in Fig. 6. As expected, the RGB visualization generated by the proposed QRCODE outperforms competing methods in terms of visual quality. It is noteworthy that the pseudo-RGB HSR HSI reconstruction using BIM [17] appears excessively smooth, even though it demonstrates robustness to misalignment cases. Furthermore, the spectral signature recovery of the HSR HSI reconstructed by methods such as USDN [33], U2MDN [34], DBIN [29], and CO-CNMF [3] is found to be suboptimal, while the results from CuCaNet [27], PZRes-Net [28], MFH-Net [31], and HySure [19] lack the fine spatial details in the reconstructed images. In contrast, the QRCODE method exhibits commendable performance in both spectral and spatial resolution.

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TABLE II

PERFORMANCE COMPARISON OF THE PROPOSED QRCODE AND THE STATE-OF-THE-ART FUSION METHODS FOR FOUR BANDS HSR MSI AND LSR HSI FUSION

Method	Aligned				1-pixel misaligned				2-pixel misaligned				3-pixel misaligned			
	PSNR↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SAM↓	RMSE↓	ERGAS↓
QRCODE	35.361	1.623	32.711	2.027	**35.229**	**1.646**	**33.111**	**2.076**	**34.957**	**1.694**	**33.936**	**2.167**	**34.599**	**1.768**	**35.253**	**2.314**
Convex Model	21.176	12.369	312.246	25.798	21.097	12.489	313.047	26.14	21.18	12.391	297.904	24.838	21.111	12.534	311.601	26.611
MHF-Net [31]	32.672	2.392	41.558	3.703	28.571	3.451	76.155	5.073	25.462	4.757	117.872	6.884	23.74	5.71	147.688	8.34
PZRes-Net [28]	34.963	1.934	35.498	1.935	32.436	2.343	48.111	2.499	29.603	2.952	68.319	3.413	27.378	3.59	90.118	4.382
CUCaNet [27]	28.848	4.14	71.71	5.035	26.585	4.458	88.605	4.158	24.537	5.034	118.992	5.08	22.724	6.345	155.868	5.893
USDN [33]	30.069	3.688	93.408	6.612	29.994	3.764	73.554	4.374	29.296	4.92	94.115	5.626	28.77	4.654	109.079	6.089
U2MDN [34]	30.127	3.235	59.071	3.725	30.081	3.15	58.023	3.679	29.9	3.245	60.536	3.827	29.814	3.285	62.989	3.817
DBIN [29]	**35.998**	**1.604**	**31.022**	**1.822**	29.695	2.493	71.177	3.366	25.271	3.745	122.266	5.529	22.787	4.813	164.46	7.333
HySure [19]	28.9836	3.3197	69.967	4.2099	26.461	4.3214	96.7679	5.7101	25.4975	4.8155	110.7174	6.3628	25.3555	4.6088	114.591	6.2021
BIM [17]	29.9873	2.7975	65.6575	3.3785	29.985	2.797	65.6444	3.3788	29.9907	2.7954	65.6087	3.3786	29.9905	2.7969	65.6227	3.3776
CO-CNMF [3]	32.4501	2.5207	45.1602	2.8059	29.0287	3.158	72.2196	3.8305	25.8697	4.2029	110.8271	5.4669	24.0282	5.019	140.0888	6.808

TABLE III

PERFORMANCE COMPARISON OF THE PROPOSED QRCODE AND THE STATE-OF-THE-ART FUSION METHODS FOR SIX BANDS HSR MSI AND LSR HSI FUSION

Method	Aligned				1-pixel misaligned				2-pixel misaligned				3-pixel misaligned			
	PSNR↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SAM↓	RMSE↓	ERGAS↓
QRCODE	38.946	**1.148**	24.617	1.429	**38.909**	**1.157**	**24.762**	**1.433**	**38.948**	**1.157**	**24.697**	**1.397**	**38.848**	**1.175**	**24.962**	**1.42**
Convex Model	32.17	6.665	125.352	4.49	32.275	6.611	122.534	4.125	32.387	6.497	116.601	3.846	32.275	6.6	119.677	3.98
MHF-Net [31]	36.116	1.801	30.633	2.947	31.44	2.879	68.716	4.249	28.344	4.101	110.384	5.963	26.636	4.98	140.392	7.274
PZRes-Net [28]	36.602	1.934	34.467	1.565	32.602	2.279	54.413	2.294	28.818	2.895	85.242	3.494	26.287	3.491	115.025	4.67
CUCaNet [27]	35.509	2.205	38.973	1.368	30.779	3.301	67.241	2.149	28.789	4.012	92.579	3.9	25.99	5.398	124.885	5.377
USDN [33]	35.208	2.65	53.987	2.307	34.814	2.304	69.194	2.937	34.428	3.036	70.599	4.193	33.841	3.433	91.214	3.527
U2MDN [34]	33.356	2.243	41.528	2.575	33.031	2.283	43.909	2.573	33.4	2.298	44.524	2.551	32.951	2.334	46.698	2.783
DBIN [29]	**39.68**	1.185	**23.693**	**1.106**	32.335	2.075	65.873	2.303	27.69	3.195	114.745	4.051	25.013	4.125	153.931	5.851
HySure [19]	31.1624	2.6973	58.784	2.9033	28.5665	3.5357	83.0885	3.9491	27.5073	3.9919	97.0628	4.5017	27.26	3.6351	102.3091	4.5278
BIM [17]	32.3671	2.0799	58.0315	2.2364	32.3741	2.0771	57.9087	2.2336	32.3702	2.078	57.9289	2.2352	32.3716	2.0778	57.9729	2.235
CO-CNMF [3]	35.7267	1.8077	33.1703	1.7961	31.1709	2.5227	62.7505	2.7722	27.5603	3.6327	101.9141	4.2787	25.6127	4.4216	130.4353	5.4311

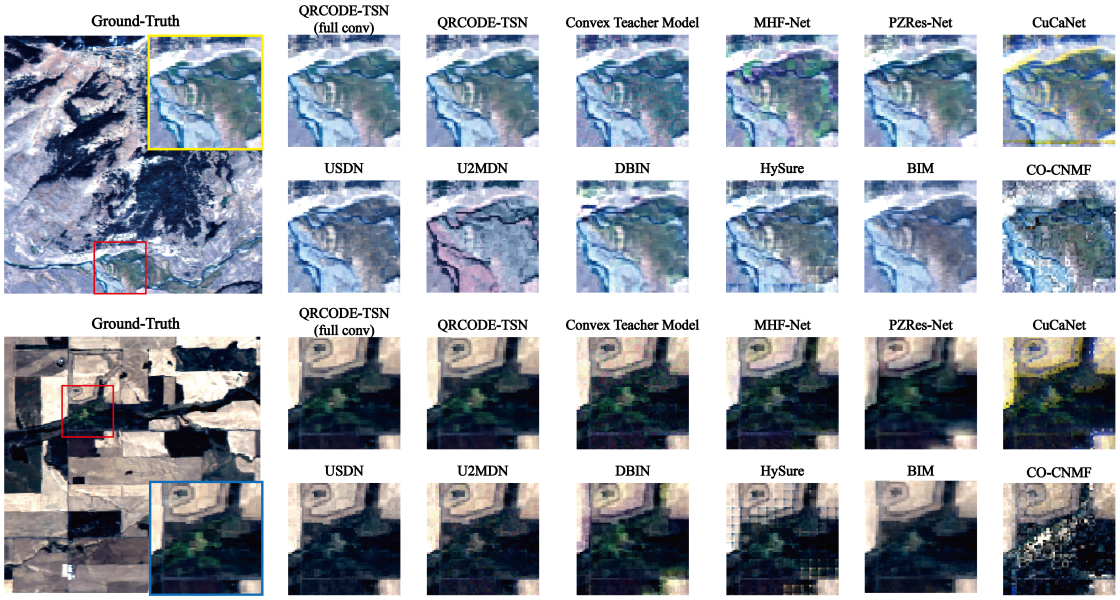

Fig. 6. RGB-visualized results of the fused HSR HSI based on four/six bands of HSR MSI and 3-pixel misaligned LSR HSI using our QRCODE, QRCODE (full conv), and other state-of-the-art peer methods.

A subjective quality comparison of RGB-visualized results under various misalignment scenarios is also presented in Fig. 7. To augment the visualization effect, the difference image between the ground truth and the reconstructed HSR HSI is calculated using $|\boldsymbol{Z}^*_{(\mathrm{RGB})} - \boldsymbol{Z}_{(\mathrm{RGB})}| + 128$, where RGB represents the false color counterparts. Although the visual quality of all methods in the aligned case is outstanding, the superiority of the proposed QRCODE becomes more pronounced as the number of misaligned pixels increases. By employing the QRCODE method, the reconstructed HSR HSI demonstrates high fidelity in both spectral and spatial domains, even under 3-pixel misaligned LSR MSI conditions.

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TABLE IV
ABLATION STUDY OF THE PROPOSED QRCODE

Convex Model	Mixed Training	Proposed SWAM	PSNR↑	SAM↓	RMSE↓	ERGAS↓
			33.630	1.875	37.552	3.214
✔			34.386	1.764	35.549	2.420
✔	✔		34.932	1.692	33.946	2.203
✔	✔	✔	**35.229**	**1.646**	**33.111**	**2.076**

TABLE V
RED HYPERPARAMETERS SELECTION OF THE PROPOSED QRCODE

Hyper parameter	Value	PSNR↑	SAM↓	RMSE↓	ERGAS↓
c_g	2	**35.603**	**1.614**	**32.264**	**1.939**
	4	35.361	1.623	32.711	2.027
	8	35.343	1.650	32.740	2.052
	10	35.369	1.642	32.768	2.051
α	10^{-1}	31.965	1.938	48.657	3.141
	10^{-2}	35.224	1.661	32.691	2.102
	10^{-3}	35.330	1.650	**32.505**	2.046
	10^{-4}	35.301	1.654	32.541	2.055
	10^{-5}	**35.361**	**1.623**	32.711	**2.027**

TABLE VI
COMPARISONS OF MODEL SIZE AND COMPLEXITY. NOTE THAT THE METHODS MARKED WITH AN ASTERISK (*) ARE UNSUPERVISED APPROACHES; M AND G INDICATE 10^6 AND 10^9; THE 4/6 BANDS CASES REFER TO THE USE OF $\boldsymbol{Y}_{m4}/\boldsymbol{Y}_{m6}$, RESPECTIVELY

Methods	Bands	#Param.	FLOPs	Run-time (in sec.)
QRCODE (full conv)	4	41M	2,231G	0.33 (1.45)
	6	41M	2,231G	0.34 (1.45)
QRCODE	4	12M	737G	0.24 (1.36)
	6	14M	812G	0.24 (1.36)
MHF-Net [31]	4	45M	4,826G	1.98
	6	91M	10,876G	3.24
PZRes-Net [28]	4	40M	5,262G	0.14
	6	40M	5,263G	0.16
DBIN [29]	4	7M	1,553G	0.19
	6	7M	1,556G	0.20
*CUCaNet [27]	4	3M	40G	2070
	6	3M	41G	2143.8
*USDN [33]	4	0.006M	1G	28.83
	6	0.006M	1G	32.33
*U2MDN [34]	4	0.01M	4G	547.28
	6	0.01M	5G	569.44

In conclusion, the QRCODE achieves the state-of-the-art performance in terms of both subjective and objective evaluations.

C. Ablation Study

1) Importance of the Proposed Modules: In Table IV, we exhibit the efficacy of each component in our proposed QRCODE by assessing its performance on a 1-pixel misaligned LSR HSI with four bands. The guidance from the convex model not only enhances the performance of our QRCODE within identical training iterations but also underscores the significance of the model's convex guidance over merely enhancing network architecture. This highlights that obtaining optimal performance is not simply reliant on intensified network structures but can be substantially improved with effective guidance, allowing for the maintenance of model simplicity.

This is evident as the model, with convex guidance, shows significant enhancement in spectral fidelity criterion (i.e., SAM), proving that including CO accelerates the achievement of superior results. On the contrary, utilizing mixed training improves performance, especially when dealing with misaligned HSR MSI and LSR HSI. However, without the supervision of the proposed SWAM, the mixed training strategy, albeit a beneficial form of data augmentation, faces challenges in learning a truly robust model.

Our main contribution is demonstrating that adequate performance can be retained under spatial misalignments. Incorporating our proposed SWAM distinctly elevates the robustness of the fusion, especially under misalignment conditions, by enabling the model to extract accurate features from misaligned HSI/MSI pairs, thus optimizing overall performance. It is worth noting that while utilizing misaligned data for training can yield performance enhancement, the evident enhancement through SWAM sheds light on its capability to learn and adapt from misaligned data effectively.

2) Hyperparameters Selection: In our QRCODE, grouped convolution is employed to efficiently manage convolutional operations by dividing them into groups. This approach significantly reduces the number of parameters and computational complexity, which is beneficial during inference. In Table V, different settings of c_g are evaluated in aligned case and indicated that a setting of $c_g = 4$ offers an optimal trade-off between model accuracy and computational efficiency. The model achieves a high PSNR in this setting, closely approaching the peak value observed at $c_g = 2$, while maintaining satisfactory SAM, RMSE, and ERGAS metrics. This balance is crucial for practical applications in constrained environments.

On the other hand, α is a key regularization term, balancing data fidelity and smoothness for HSI/MSI fusion tasks. Through a detailed examination of various α values, we identified $\alpha = 10^{-5}$ as the optimal setting, as shown in Table V. This choice demonstrated superior performance in critical metrics such as SAM and ERGAS, indicating high spectral fidelity and overall accuracy in HSI reconstruction. Although there is a slight increase in RMSE at $\alpha = 10^{-5}$ compared to $\alpha = 10^{-3}$, the improvements in spectral fidelity and overall error justify its selection for our application, emphasizing the model's ability to achieve high spectral fidelity and low overall error in practical hyperspectral imaging applications.

D. Robustness Evaluation

Spatial misalignment in HSI/MSI can manifest in multiple forms. For example, when acquiring data from mobile platforms like aircraft and satellites, fluctuations in camera motion can introduce nonrigid distortions owing to alterations in orientation, velocity, and vibrations. These distortions are exacerbated in regions with diverse perspectives, impacting spatial accuracy. Moreover, atmospheric elements, such as variations in temperature, humidity, and airflow, can precipitate light refraction and scattering, inducing nonlinear deformations in the captured images.

Simulating nonrigid distortions is also essential. It enables a comprehensive evaluation of the robustness and efficacy of

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Fig. 7. RGB-visualized and different results of the fused HSR HSI based on four bands of HSR MSI and different misalignment factors of LSR HSI using our QRCODE, QRCODE (full conv), and other state-of-the-art peer methods.

various image registration and reconstruction methodologies. A possible approach for simulating these distortions involves employing a bimodal Gaussian function to mimic elastic deformations in images, a phenomenon frequently observed in the real world due to various physical and environmental influences. The bimodal Gaussian function is defined as the summation of two Gaussian functions, each described by its mean μ and standard deviation σ, and can be expressed as

$$G(x) = G_1(x; \mu_1, \sigma_1) + G_2(x; \mu_2, \sigma_2). \tag{23}$$

Each pixel at position (i, j) in an image undergoes deformations computed using the bimodal Gaussian function, resulting in deformation values Δx and Δy. The shifted pixel positions

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TABLE VII
PERFORMANCE COMPARISON OF THE PROPOSED QRCODE EVALUATED IN LINEAR AND NONRIGID MISALIGNMENT TESTING DATASET. THE RED AND BOLD FONT REPRESENTS THE FIRST POSITION IN THE EVALUATION METRICS, WHILE THE BLUE FONT INDICATES THE SECOND POSITION

Testing distortion	Shifted pixels	PSNR ↑	SAM ↓	RMSE ↓	ERGAS ↓
Linear	0	**35.361**	**1.623**	**32.711**	**2.027**
	1	35.229	1.646	33.111	2.076
	2	34.957	1.694	33.936	2.167
	3	34.599	1.768	35.253	2.314
Non-rigid	$\beta = 1$	35.288	1.634	32.907	2.046
	$\beta = 2$	35.168	1.654	33.265	2.087
	$\beta = 3$	35.003	1.676	33.660	2.137

are then calculated by

$$(i', j') = \beta(i + \Delta x, j + \Delta y) \tag{24}$$

where β represents the degree of distortion. It can be regarded as a shifted pixels operation in linear misalignment. For this experiment, the parameters are set as follows: $\mu = 1.5$, $\sigma = 1.0$, and $\beta \in [1, 3]$. As the modified pixel positions (i', j') do not align with the integer pixel values of the original image, an appropriate interpolation technique, such as bilinear or cubic interpolation, is essential for estimating pixel values at the new locations. In the subsequent experiments, bilinear interpolation is employed to estimate the pixel values in the nonrigidly distorted images.

Table VII presents the performance comparison of the proposed QRCODE, tested on aligned, linearly misaligned, and nonrigidly misaligned HSI/MSI pairs. Although our QRCODE model is fundamentally trained on linearly misaligned data, it exhibits promising performance under nonrigid distortions. This implies that our SWAM is effective and robust in managing spatial distortions arising from both nonrigid and linear misalignments. We also assess the performance of our QRCODE when trained on nonrigidly misaligned data, and the outcomes remain commendable. Considering the negligible performance gap between linear and nonrigid distorted pairs, we argue for utilizing the linearly misaligned pair as the training set for our QRCODE, thereby optimizing efficiency during the training phase.

E. Computational and Space Complexity Comparison

In this article, a comprehensive evaluation of computational and space complexity is conducted between the proposed QRCODE method and other DL-based techniques for HSI and MSI data fusion. A fair comparison is ensured by maintaining identical input sizes and batch sizes across all models, as detailed in Table VI. Note that two run-time records for the proposed QRCODE are measured, in which the runtime of our QRCODE with the CO solution is denoted in brackets.

By employing grouped convolution in the QRCODE model, the reduction of parameters and FLOPs is achieved by 70% and 67%, respectively. This highlights the efficient utilization of spectral redundancy between HSI and MSI data through grouped convolution, ultimately contributing to the favorable outcomes obtained using the QRCODE model. Moreover, a significant improvement in inference time is achieved due to the reduced FLOPs.

Although the DBIN method [29] demonstrates a relatively lower number of parameters, its computational complexity, as quantified by FLOPs, remains higher than that of the proposed QRCODE, attributed to its intricate network architecture. Furthermore, the QRCODE model significantly surpasses DBIN [29] in the HSI/MSI fusion task, particularly in cases involving misalignments. Unsupervised learning-based approaches, such as CUCaNet [27], USDN [33], and U2MDN [34], exhibit lower space and computational complexities in their network designs. However, they necessitate online training for HSI/MSI fusion, resulting in extended inference times.

In conclusion, the proposed QRCODE method attains comparable run-time efficiency relative to alternative approaches. Additionally, it achieves the state-of-the-art performance in HSI/MSI fusion tasks, even under misalignment scenarios. Owing to the rapid solution of the CO-based FD; the QRCODE method establishes an optimal balance between performance and efficiency.

IV. CONCLUSION

In this article, we propose a novel SWAM-integrated QRCODE algorithm, which seamlessly combines the merits of CO and DL, to effectively address the misaligned HSI/MSI fusion without performance degradation. The primary objective is to economically obtain HSR HSIs, which are crucial for accurate remote material identification and classification. By incorporating a simple and fast convex model, we facilitate the learning of the spatial-spectral relationship when guiding the quasi-residual learning task. The low-rank convex model effectively leverages the convex geometry structure in the hyperspectral signature space. The proposed convex subnetwork not only avoids the need for sophisticated network structures commonly found in conventional models but also offers enhanced stability due to its inherent convexity. By examining the spectral redundancy between the LSR HSI and HSR MSI, we judiciously employ grouped convolution in the proposed fast alignment-free network, reducing computational and space complexities without compromising performance. Integrating the SWAM further enhances the performance of the proposed QRCODE algorithm in addressing misaligned HSI/MSI fusion scenarios. Experimental results validate the superiority of the proposed QRCODE, incorporating SWAM, over other benchmark methods, particularly in misaligned situations.

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